



DEALER BEST PRACTICE SERIES

Cleaning and Recycling of Metallic Waste in Machining Shop

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

Cleaning and Recycling of Metallic Waste in Machining Shop	0
1.0 Introduction	1
2.0 Best Practice Description.....	1
3.0 Implementation Steps	2
4.0 Benefits.....	5
5.0 Resources Required	5
6.0 Supporting Attachments / References	5
7.0 Related Best Practices	5
8.0 Acknowledgements.....	5

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

1.0 Introduction

New cleaning equipment which improves the control, removal, and recycling of metallic waste has been implemented in the machine shop in Ferreyros SA. The equipment is an industrial vacuum.

2.0 Best Practice Description

When large components (salvaged by welding) are machined, a large amount of metallic metal shavings are created. This residue hinders the work of the operators, poses a contamination control risk, and creates an unprofessional shop image.



Picture 1. Metal shavings on floors and tooling.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Cleaning and Recycling of Metallic Waste in Machining Shop	DATE 2/18/2015	CHG NO 00	NUMBER 1406-4.02-1343
--	-------------------	-----------------	--------------------------



Picture 2. Some Large Components must be cleaned mid way through the machining in order to continue the process.

3.0 Implementation Steps



Picture 3. The solution that has been implemented is the use of a large, industrial quality vacuum that is easily portable in the shop

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Cleaning and Recycling of Metallic Waste in Machining Shop	DATE 2/18/2015	CHG NO 00	NUMBER 1406-4.02-1343



Picture 4. Cleaning ERS Dipper Saddle



Picture 5. Recycling of metallic waste

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Cleaning and Recycling of Metallic Waste in Machining Shop	DATE	CHG NO	NUMBER
	2/18/2015	00	1406-4.02-1343

Features of the industrial vacuum:

- Trademark: WIELAND
- Model: IS-46M ATEX 3D
- Weight : 143 Kg
- Volt: 230
- Kw: 478
- Max mains prefuse: 25 amp

EDITOR'S NOTE: *Although Caterpillar does not endorse this particular supplier, Ferreyros has experience with the above products or companies. We welcome additional supplier recommendations from our dealer network.*

- Before and after -



Picture 6. Cleaning of metallic waste

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Cleaning and Recycling of Metallic Waste in Machining
Shop

DATE
2/18/2015

CHG
NO
00

NUMBER
1406-4.02-1343

4.0 Benefits

- Reduce cleaning time by 80%
- Reduce the number of staff for cleaning (from 2 people to 1)
- Better monitoring of metallic waste
- It is portable and can be used in any work area that has power output.

5.0 Resources Required

Purchase cost of a industrial vacuum.

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

Team: Miguel Mauriola / Operations Support Chief
Romulo Olarte /Supervisor
José Huayamis / Technical Communicator
Owner: Rafael Volta / Machine Shop Chief
Sponsor: Rodolfo Gerstein / Manager Machine Shop

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Cleaning and Recycling of Metallic Waste in Machining Shop	DATE 2/18/2015	CHG NO 00	NUMBER 1406-4.02-1343
--	-------------------	-----------------	--------------------------